The WALE Subgrid-Scale Model in Large Eddy Simulation: Motivation, Near-Wall Behaviour, and Comparison with Alternative SGS Models

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Abstract

The wall-adapting local eddy-viscosity (WALE) model is one of the most influential subgrid-scale (SGS) closures used in large eddy simulation (LES) of wall-bounded turbulent flows. This literature review examines the motivation, formulation, advantages, limitations, and later developments of the WALE model within the broader context of SGS modelling for LES. The review begins with the classical Smagorinsky model and the dynamic Smagorinsky model as important precursors, emphasizing the challenges of near-wall and transitional-flow modelling that motivated the development of WALE. The formulation of WALE is then presented in detail, with emphasis on its use of the square of the resolved velocity-gradient tensor, its sensitivity to both strain and rotation, and its ability to recover the correct near-wall cubic scaling of the eddy viscosity without requiring an explicit damping function. The review then assesses the model's performance, including its behaviour in near-wall, transitional, and wall-bounded turbulent flows, before discussing its main limitations, such as its fixed model constant, non-zero response in solid-body rotation, and less consistent behaviour in some under-resolved LES settings. Later developments motivated by these limitations, including dynamic WALE formulations and the σ-model, are also examined. Although WALE is not universally optimal, the review concludes that it remains an important SGS model both in its own right and as a foundation for later robust wall-adapting closures.

1. Introduction

Turbulent flows remain one of the central challenges in computational fluid dynamics because the flow features range from large energy-containing motions to very small dissipative scales. In principle, the most complete numerical description of turbulence is provided by direct numerical simulation (DNS), in which all relevant scales of motion are resolved without modelling. In practice, however, the computational cost of DNS skyrockets as the Reynolds number increases. For homogeneous isotropic turbulence, the number of grid points required scales as $Re_T^{9/4}$, while the total computational effort scales approximately as $Re_T^{11/4}$ once the timestep restriction is taken into account (Coleman & Sandberg 2010). As a result, DNS is not a practical tool for most engineering flows, particularly those involving high Reynolds numbers, wall-bounded turbulence, or complex geometries. This limitation motivates the use of a reduced-order approach that preserves the physics of turbulence at a lower computational cost. LES addresses this need by resolving the larger energy-containing eddies, while the effects of the smaller unresolved scales are represented through an SGS model (Pope 2000).

The success of LES depends strongly on the choice of SGS model, particularly in wall-bounded flows where near-wall turbulence is difficult to represent accurately. Among the

SGS closures proposed, the WALE model has become one of the most widely discussed because of its improved near-wall behaviour and its relatively simple algebraic form.

The objective of this literature review is to examine the motivation, development, formulation, and performance of the WALE model within the broader context of SGS modelling for LES. The review begins by introducing the general background of SGS modelling and then discusses the limitations of earlier SGS closures. It next presents the formulation of the WALE model, followed by an evaluation of its main advantages and current limitations. Finally, the review considers variations of the WALE model and outlines possible directions for future research.

2. Background

The application of LES begins with the spatial filtering of the Navier-Stokes equations. This introduces the SGS stress tensor, defined as

$$\tau_{ij} = \overline{u_i u_j} - \bar{u}_i \bar{u}_j, \tag{2.1}$$

which must be modelled to close the filtered equations (Germano *et al.* 1991). A common approach is to use an eddy-viscosity assumption, in which the anisotropic part of the SGS stress is written as

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2\nu_t \bar{S}_{ij}, \tag{2.2}$$

where ν_t is the SGS eddy viscosity and $\bar{S}_{ij}$ is the resolved strain-rate tensor,

$$\bar{S}_{ij} = \frac{1}{2}\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right). \tag{2.3}$$

This framework is particularly attractive because of its simplicity and relatively low computational cost. Instead of attempting to reconstruct the exact unresolved stresses, it models their overall dissipative effect on the resolved scales.

The main difficulty is that the unresolved motions are not represented equally well in all parts of the flow. In particular, near solid walls, the turbulence becomes highly anisotropic, and the relevant scales become much smaller. As a result, SGS models that rely too directly on the local strain rate may produce excessive eddy viscosity and over-damp the resolved flow (Germano *et al.* 1991; Nicoud & Ducros 1999). The issue is one of the central challenges in LES of wall-bounded turbulence and strongly motivates the development of improved SGS closures.

3. Smagorinsky and Dynamic Smagorinsky Models

This section reviews the Smagorinsky model and the dynamic Smagorinsky model as important precursors to the WALE formulation. Their strengths and weaknesses help explain the motivation for later near-wall SGS closures.

3.1. *Smagorinsky Model*

One of the earliest and most widely used SGS closure is the Smagorinsky model, in which the eddy viscosity is given by

$$\nu_t = (C_s \Delta)^2 |\bar{S}|, \tag{3.1}$$

where

$$|\bar{S}| = \left(2\bar{S}_{ij}\bar{S}_{ij}\right)^{1/2}. \tag{3.2}$$

Here, C_s is the Smagorinsky constant and Δ is the filter width (Smagorinsky 1963). This model is attractive because it is simple, local, inexpensive, and robust. As noted by Germano et al., for homogeneous isotropic turbulence, Lilly estimated $C_s \approx 0.17$ by matching the Smagorinsky model to the inertial-range predicted by the Kolmogorov theory (Germano *et al.* 1991).

Despite these advantages, the Smagorinsky model has important limitations. One of which is that it performs poorly in transitional flows, where mean shear can trigger excessive dissipation and suppress the physical growth of instabilities. In addition, the Smagorinsky model uses a fixed coefficient and therefore cannot adapt to different flow regimes or represent backscatter in any natural way. Its most important weakness, however, appears near a no-slip wall. Because ν_T depends directly on the local strain-rate magnitude, it remains non-zero wherever strong velocity gradients exist. Near a no-slip wall, this causes excessive eddy viscosity and over-dissipation of the resolved motion, even though turbulence fluctuations should vanish at the wall. Hence, early LES studies often introduced ad hoc damping functions, such as Van Driest damping, to reduce the model near solid boundaries. However, these corrections are forced, depend explicitly on wall distance, and are less attractive in complex geometries (Germano *et al.* 1991; Nicoud & Ducros 1999). These limitations strongly motivated the development of improved SGS closures (Germano *et al.* 1991; Nicoud & Ducros 1999).

3.2. *Dynamic Smagorinsky Model*

A major improvement was introduced by Germano et al. through the Dynamic Smagorinsky model, where the model coefficient is not prescribed a priori but is instead computed from the resolved flow and the simulation evolves. The key idea is to apply a second, larger "test" filter on top of the grid filter. Referring back to the grid-filter SGS stress in equation (3.1), the SGS stress at the test-filter is

$$T_{ij} = \widetilde{\bar{u}_i\bar{u}_j} - \widetilde{\bar{u}}_i\widetilde{\bar{u}}_j, \tag{3.3}$$

and the resolved turbulent stress between the two filter is

$$L_{ij} = \widetilde{\bar{u}_i\bar{u}_j} - \widetilde{\bar{u}}_i\widetilde{\bar{u}}_j. \tag{3.4}$$

These quantities are linked through the Germano identity,

$$L_{ij} = T_{ij} - \widetilde{\tau}_{ij}. \tag{3.5}$$

Assuming a Smagorinsky form for both filter levels, the anisotropic part of the SGS stress is modelled as

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2C\Delta^2|\bar{S}|\,\bar{S}_{ij}, \tag{3.6}$$

and similarly at the test-filter level,

$$T_{ij} - \frac{1}{3}T_{kk}\delta_{ij} = -2C\widetilde{\Delta}^2|\widetilde{\bar{S}}|\,\widetilde{\bar{S}}_{ij}, \tag{3.7}$$

Where $C = C_s^2$ is now treated as a variable rather than a universal constant. Germano

et al. then used the resolved stress relation to determine C dynamically from the local or averaged flow field. In practice, Lilly later recast this into a least-squares form to avoid over-determination (Germano *et al.* 1991).

The dynamic Smagorinsky model represented a major advance over the original model because it allowed the Smagorinsky constant to adapt to the instantaneous resolved flow rather than being prescribed as a universal value. As a result, the model reduced the excessive eddy viscosity produced by the Smagorinsky model in laminar, transitional, and near-wall regions. It was shown that the dynamic formulation yield zero SGS stress in laminar flow and at solid boundaries, recovers the correct near-wall asymptotic behaviour without ad hoc damping functions, and does not exclude the possibility of backscatter. They also reported good agreement with DNS in LES transitional and turbulent channel flow (Germano *et al.* 1991).

However, the dynamic Smagorinsky model is not perfect. It requires an additional filtering operation at the test-filter scale, and in its classical implementation, the dynamically computed coefficient often needs averaging in homogeneous directions to avoid numerical instability and large negative values. This makes the method harder to apply in complex geometries or fully inhomogeneous flows. These limitations were an important motivation for models such as WALE, which seek improved near-wall behaviour without explicit damping functions or dynamic test filtering.

4. The WALE model

The WALE model, introduced by Nicoud and Ducros in 1999, was developed to address key shortcomings of earlier SGS closures, particularly their poor behaviour near no-slip walls. The main goal of the model is to provide an SGS eddy viscosity that vanishes naturally at the wall without requiring an explicit damping function. In particular, the model was designed to recover the correct near-wall asymptotic behaviour,

$$\nu_t = O(y^3) \qquad \text{as } y \to 0,$$

which is regarded as an essential requirement for SGS modelling in wall-bounded LES. To achieve this, the WALE model defines the eddy viscosity using a quantity derived from the square of the resolved velocity-gradient tensor rather than from the resolved strain rate alone. This allows the model to remain local, simple, and based only on resolved flow quantities, while also accounting for both strain and rotation effects of the smallest resolved motions. As a result, the WALE model has become one of the most widely discussed SGS closures for LES of wall-bounded turbulence.

4.1. *Formulation of the WALE model*

As discussed above, the WALE model was designed to construct an eddy viscosity with the correct near-wall behaviour while remaining local and based only on resolved quantities. In this framework, the SGS eddy viscosity is assumed to take the general form

$$\nu_t = C\Delta^2\overline{OP(x,t)}, \tag{4.1}$$

where C is a model constant, Δ is the local filter width, and $\overline{OP(x,t)}$ is an operator built from the resolved velocity field.

To define this operator, let the resolved velocity-gradient tensor be

$$g_{ij} \equiv \frac{\partial \bar{u}_j}{\partial x_i}. \tag{4.2}$$

Unlike the Smagorinsky model, which depends only on the symmetric part of this tensor through the strain-rate tensor $\bar{S}_{ij}$ introduced in Eq. (2.3), the WALE model is based on the full velocity-gradient tensor. This tensor may be decomposed as

$$g_{ij} = \bar{S}_{ij} + \Omega_{ij}, \tag{4.3}$$

where

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial \bar{u}_j}{\partial x_i} - \frac{\partial \bar{u}_i}{\partial x_j}\right) \tag{4.4}$$

is the resolved rotation-rate tensor. In this way, the WALE model incorporates information from both strain and rotation, rather than strain alone.

The key ingredient in the WALE formulation is the traceless symmetric part of the square of the velocity-gradient tensor,

$$S^d_{ij} = \frac{1}{2}\left(g_{ik}g_{kj} + g_{jk}g_{ki}\right) - \frac{1}{3}\delta_{ij}g_{mn}g_{nm}. \tag{4.5}$$

Because S^d_{ij} is formed from the full tensor g_{ij}, it reflects both strain and rotational effects of the smallest resolved motions.

The operator in Eq. (4.1) is then written as

$$OP(x,t) = \frac{OP_1}{OP_2}, \tag{4.6}$$

with

$$OP_1 = \left(S^d_{ij}S^d_{ij}\right)^{3/2}, \tag{4.7}$$

and

$$OP_2 = \left(\bar{S}_{ij}\bar{S}_{ij}\right)^{5/2} + \left(S^d_{ij}S^d_{ij}\right)^{5/4}. \tag{4.8}$$

Thus, the WALE eddy viscosity becomes

$$\nu_t = (C_w\Delta)^2 \frac{\left(S^d_{ij}S^d_{ij}\right)^{3/2}}{\left(\bar{S}_{ij}\bar{S}_{ij}\right)^{5/2} + \left(S^d_{ij}S^d_{ij}\right)^{5/4}}. \tag{4.9}$$

An important reason for using this operator is that $S^d_{ij}S^d_{ij}$ does not treat shear alone as a sufficient indicator of SGS activity. Expanding S^d_{ij} in terms of the strain-rate and rotation-rate tensors shows that

$$S^d_{ij}S^d_{ij} = \frac{1}{6}\left(\bar{S}_{ij}\bar{S}_{ij}\,\bar{S}_{ij}\bar{S}_{ij} + \Omega_{ij}\Omega_{ij}\,\Omega_{ij}\Omega_{ij}\right) + \frac{2}{3}\Omega_{ij}\Omega_{ij}\,\bar{S}_{ij}\bar{S}_{ij} + 2\,\bar{S}_{ik}\bar{S}_{kj}\Omega_{j\ell}\Omega_{\ell i}. \tag{4.10}$$

Nicoud and Ducros showed that, for pure shear, this quantity vanishes,

$$S^d_{ij}S^d_{ij} = 0, \tag{4.11}$$

so the WALE model does not generate eddy viscosity solely because a local shear is present. This is one of the reasons it behaves more appropriately than the Smagorinsky model in near-wall regions.

In practice, the filter width is commonly taken as the cube root of the cell volume,

$$\Delta = (\Delta x\,\Delta y\,\Delta z)^{1/3}, \tag{4.12}$$

and the constant is often taken as $C_w \approx 0.325$ (Nicoud & Ducros 1999).

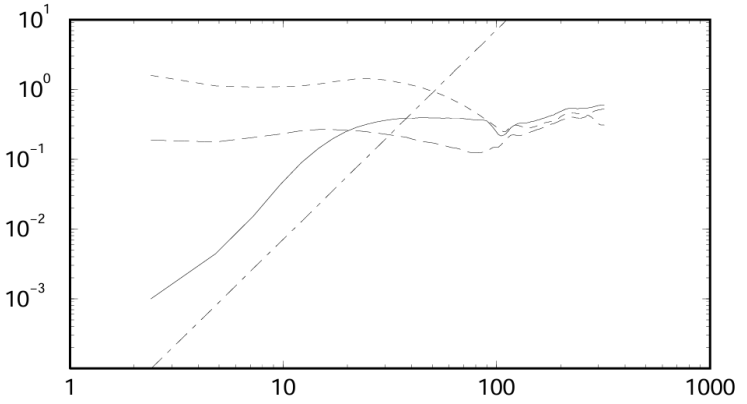


Figure 1: Ratio of average eddy viscosity to molecular viscosity versus distance from the wall in log–log coordinates. Shown are results for the classical Smagorinsky model (dotted line), the filtered Smagorinsky model (dashed line), and the WALE model (solid line). The dot-dashed line denotes the proper near-wall scaling. The WALE model follows the expected near-wall behaviour much more closely than the Smagorinsky-based formulations. Adapted from Nicoud and Ducros (1999).

4.2. *Advantages and Performance of the WALE model*

An important advantage of the WALE model is that it is sensitive to both strain and rotation, rather than to the strain rate alone. This gives it a broader physical basis than the classical Smagorinsky model, whose eddy viscosity depends only on the strain-rate magnitude. By incorporating information from the full resolved velocity-gradient tensor, the WALE model is better suited to represent local SGS activity in flows where both strain and rotational motions are important (Nicoud & Ducros 1999).

A second major advantage is its near-wall behaviour. Because the WALE eddy viscosity decreases with the cube of the wall-normal distance, it vanishes naturally at a no-slip wall without requiring an explicit damping function. This is one of its main improvements over the Smagorinsky model, whose near-wall response is generally too large unless additional corrections are introduced.

This behaviour is also evident in Fig. 1, where the WALE curve follows the expected near-wall trend much more closely than the classical and filtered Smagorinsky models (Nicoud & Ducros 1999).

The model also has practical advantages for LES in complex geometries. Unlike the dynamic Smagorinsky model, WALE is fully local and does not require test filtering, plane averaging, or explicit dependence on wall distance. It is invariant under coordinate translations and rotations, depends only on local resolved flow quantities, and can be implemented on both structured and unstructured grids (Nicoud & Ducros 1999). These features make it especially attractive in configurations where homogeneous directions are absent or wall distance is not straightforward to define.

Another useful feature of the WALE model is its behaviour in transitional flows. Because the WALE operator vanishes in pure shear, it does not produce excessive eddy viscosity in laminar shear regions. This allows unstable modes to grow more naturally and improves the model's ability to reproduce laminar-to-turbulent transition. In the turbulent pipe-flow study of Nicoud and Ducros, the WALE model was able to sustain transition and reach a statistically steady turbulent state, whereas the Smagorinsky model tended to over-damp the flow in such situations (Nicoud & Ducros 1999).

Nicoud and Ducros also tested the model in freely decaying isotropic turbulence and reported that it produced an appropriate level of dissipation when compared with the experimental spectra of Comte-Bellot and Corrsin. In their turbulent pipe-flow calculations, the model reproduced the expected near-wall scaling of eddy viscosity, captured the logarithmic mean velocity profile reasonably well, and gave turbulence intensities in fair agreement with the available data. They also found that, near the wall, the eddy viscosity predicted by WALE remained much smaller than that produced by Smagorinsky-type models (Nicoud & Ducros 1999).

These results support WALE as a substantial improvement over earlier strain-based SGS closures for wall-bounded LES. At the same time, the original work also noted that the model should be tested further in more complicated flow configurations and that its extension to compressible flows is not entirely straightforward (Nicoud & Ducros 1999).

4.3. *Limitations of the WALE model*

Despite its advantages, the WALE model also has several limitations. First, it is somewhat more expensive than the classical Smagorinsky model because it requires the evaluation of the square of the resolved velocity-gradient tensor rather than only the strain-rate tensor. Although this additional cost is usually modest, it makes the model algebraically more complex than simpler strain-based closures.

A second limitation is that, in its standard form, WALE uses a fixed model constant C_w. As a result, it cannot automatically adapt to changes in local flow conditions, grid resolution, or numerical error. Like other static eddy-viscosity models, this means that the SGS dissipation may still be too weak in some regions and too strong in others, depending on the flow and the level of resolution.

This issue becomes more apparent in severely under-resolved LES. For example, Duan and Wang reported that, in some wall-modelled channel-flow cases, WALE could provide insufficient dissipation and was outperformed by the Vreman model (Duan & Wang 2024). This suggests that, although WALE improves near-wall behaviour relative to Smagorinsky-type models, it is not universally optimal for all LES settings.

Another known limitation is that the WALE model does not vanish in the case of solid-body rotation. In situations where pure rotation is present without the kind of local deformation associated with SGS dissipation, this can lead to a non-zero eddy viscosity that is not always physically desirable. This point has motivated later efforts to develop alternative operators that preserve the favourable near-wall behaviour of WALE while improving its response in specific limiting flow cases (Nicoud *et al.* 2011).

5. Later developments motivated by WALE limitations

The WALE model resolved several important deficiencies of earlier SGS closures, especially in its treatment of near-wall turbulence. However, it does not eliminate all modelling difficulties. As discussed in the previous section, WALE remains a static eddy-viscosity model with a fixed coefficient, can produce a non-zero response in some limiting flow configurations such as solid-body rotation, and may provide insufficient dissipation in severely under-resolved LES. These remaining issues have motivated later developments that build on WALE in two main ways. One approach is to retain the WALE operator but adapt its coefficient dynamically, leading to dynamic WALE formulations. Another is to modify the SGS operator itself while preserving the desirable locality and near-wall behaviour of WALE, as in the later σ-model. Examining these developments helps clarify both the limitations of WALE and the reasons why its core ideas remain influential in more recent SGS modelling.

5.1. *Dynamic WALE models*

One natural way to address a limitation of the WALE model is to replace its fixed model constant with a dynamically computed coefficient. In principle, this would allow the model to adapt to changing flow conditions, grid resolution, and local levels of turbulence, in the same spirit as the dynamic Smagorinsky model. Baya, Truffin, and Nicoud developed such a formulation by applying the Germano identity to the WALE closure (Toda *et al.* 2010).

To construct the model, a second test filter of width $\hat{\Delta}$ is introduced in addition to the grid filter of width Δ. The SGS stress at the test-filter level is denoted T_{ij}^{sgs}, and the Germano identity is written as

$$T_{ij}^{sgs} - \widehat{\tau_{ij}^{sgs}} = L_{ij}, \tag{5.1}$$

where L_{ij} is the Leonard stress,

$$L_{ij} = \widehat{\bar{u}_i \bar{u}_j} - \hat{\bar{u}}_i \hat{\bar{u}}_j. \tag{5.2}$$

Using the WALE closure at both filter levels gives

$$C_w^2 \hat{\Delta}^2 \widehat{|OP|}\, \hat{\bar{S}}_{ij} - C_w^2 \widehat{\Delta^2 |OP|} \bar{S}_{ij} = L_{ij}, \tag{5.3}$$

and, assuming that C_w is approximately constant over a region at least as large as the test filter, the coefficient can be obtained using Lilly's least-squares procedure as

$$C_w^2 = \frac{1}{2} \frac{\langle L_{ij} M_{ij}^w \rangle^+}{\langle M_{ij}^w M_{ij}^w \rangle}, \tag{5.4}$$

with

$$M_{ij}^w = \hat{\Delta}^2 \widehat{|OP|}\, \hat{\bar{S}}_{ij} - \Delta^2 \widehat{|OP| \bar{S}_{ij}}. \tag{5.5}$$

Here, $\langle \cdot \rangle$ denotes a local stabilizing average and the superscript + indicates clipping of negative values to zero.

At first sight, this appears to be a straightforward extension of the dynamic Smagorinsky idea. However, Baya et al. argued that this procedure is somewhat paradoxical when applied to WALE. The original purpose of WALE was precisely to recover the correct near-wall behaviour without requiring either a damping function or a dynamic procedure. A naive dynamic WALE therefore risks damaging the very property that made the model attractive in the first place (Toda *et al.* 2010).

Their channel-flow results show that this concern is not merely conceptual. In homogeneous isotropic turbulence, the dynamic WALE model behaved similarly to the static WALE model and recovered a coefficient close to the original constant proposed by Nicoud and Ducros. In turbulent channel flow, however, the dynamic WALE coefficient became excessively large near the wall, leading to an overprediction of the eddy viscosity in the buffer layer and a poor mean-velocity prediction. Baya et al. explained this through a near-wall asymptotic analysis: because the WALE operator itself scales like y^3, the numerator and denominator in the Germano-based expression for C_w^2 both involve very small quantities near the wall. Formally, the coefficient remains of order y^0, but numerically the ratio becomes ill-conditioned and highly sensitive to the details of the filtering and stabilization procedure (Toda *et al.* 2010).

This is the central difficulty of dynamic WALE models. For a closure such as Smagorinsky, the dynamic procedure is useful because it compensates for the model's poor near-wall behaviour. For WALE, by contrast, the base model already contains the desired near-wall scaling. As a result, the Germano procedure does not simply improve the model; it can interfere with the mechanism that gives WALE its wall-adapting character. In this sense, the difficulty is not merely numerical but structural: the local dynamic procedure is least well

behaved precisely in the region where WALE is already doing something physically correct (Toda *et al.* 2010).

To address this problem, Baya et al. proposed a modified dynamic WALE model based on a shear-and-vortex sensor (SVS), which is designed to distinguish the near-wall region from the fully developed turbulent core without requiring explicit geometric information. They defined

$$SVS = \frac{(S^d_{ij}S^d_{ij})^{3/2}}{(S^d_{ij}S^d_{ij})^{3/2} + (\bar{S}_{ij}\bar{S}_{ij})^3}, \tag{5.6}$$

which behaves like y^3 near the wall, is zero in pure shear, and approaches unity in pure rotation. Using this sensor, they kept the static WALE constant in the near-wall region and evaluated the coefficient dynamically only where the flow was sufficiently far from the wall. In their implementation,

$$C^2_{w,SVS} = \frac{1}{2}\frac{\langle L_{ij}M^w_{ij}\rangle^+}{\langle M^w_{ij}M^w_{ij}\rangle} \qquad \text{if } SVS > 0.09, \tag{5.7}$$

whereas

$$C^2_{w,SVS} = 0.25 \qquad \text{if } SVS < 0.09. \tag{5.8}$$

This hybrid strategy preserved the proper near-wall behaviour while allowing dynamic adaptation in the turbulent core, and it produced substantially better channel-flow results than the naive dynamic WALE formulation (Toda *et al.* 2010).

From the perspective of this review, the main lesson is that dynamic WALE models are not a simple upgrade of the original closure. They reveal a tension between two modelling goals: local dynamic adaptation and built-in near-wall asymptotics. The work of Baya et al. shows that a straightforward Germano-type dynamic procedure can undermine WALE's main strength, but it also suggests a more constructive path forward, namely hybrid formulations that preserve the static near-wall structure of WALE while introducing dynamic adaptation only where it is actually beneficial. This idea connects naturally to later developments in WALE-based and WALE-inspired SGS modelling.

5.2. *σ-model*

A different response to the limitations of WALE is not to modify the model constant dynamically, but to change the SGS operator itself. One known weakness of WALE is that, although it preserves the correct near-wall cubic behaviour, it does not vanish in the case of solid-body rotation. This is undesirable because solid rotation, by itself, should not necessarily be interpreted as an indicator of SGS dissipation. The σ-model was proposed to address this issue while retaining the main advantages of WALE, namely locality, low computational cost, and the correct near-wall behaviour.

Instead of constructing the operator from the square of the resolved velocity-gradient tensor, the σ-model is built from the singular values of the velocity-gradient tensor g_{ij}. Let

$$\sigma_1 \geqslant \sigma_2 \geqslant \sigma_3 \geqslant 0$$

denote the three singular values of g_{ij}. The model then defines the operator

$$D_\sigma = \frac{\sigma_3(\sigma_1 - \sigma_2)(\sigma_2 - \sigma_3)}{\sigma_1^2}, \tag{5.9}$$

so that the eddy viscosity is written as

$$\nu_t = (C_\sigma \Delta)^2 D_\sigma. \tag{5.10}$$

Because this formulation depends on the ordered singular values of the velocity-gradient tensor, it remains purely local while satisfying a broader set of desirable limiting-flow properties than WALE. In particular, the σ-model vanishes in pure shear and in solid-body rotation, while still recovering the correct near-wall cubic scaling (Nicoud *et al.* 2011).

The numerical tests reported in the original paper also support the promise of the model. Nicoud et al. examined the σ-model in decaying isotropic turbulence and in periodic turbulent channel flow using several different numerical solvers. They reported results that were systematically equivalent to, or slightly better than, those obtained with the dynamic Smagorinsky model, while still preserving the simplicity and low computational cost of a local static SGS closure (Nicoud *et al.* 2011).

In this sense, the σ-model may be viewed as a refinement of the WALE philosophy rather than a complete departure from it. Both models seek to improve on Smagorinsky while preserving locality and proper near-wall behaviour, but the σ-model extends this idea by enforcing a broader set of desirable limiting-flow properties through the singular values of the velocity-gradient tensor.

5.3. *Is WALE still useful?*

Despite its limitations, the WALE model remains highly useful. Its continued relevance does not rest on the claim that it is always the best SGS closure in every LES setting, but rather on the fact that it introduced a successful modelling idea: a local eddy-viscosity operator that recovers the correct near-wall behaviour without requiring wall damping functions or dynamic test filtering (Nicoud & Ducros 1999). Even where later models outperform WALE in specific cases, its basic design has remained influential.

One reason WALE is still useful is that it often serves as a foundation for more robust SGS formulations. Rather than abandoning the WALE operator entirely, later developments have reused or modified it to retain its favourable wall-adapting properties while improving robustness, dissipation control, or behaviour in special flow configurations. In this sense, WALE should be viewed not only as a standalone model, but also as a building block for subsequent SGS closures.

A clear example is the RVM-WALE model, in which the WALE operator is incorporated into a regularized variational multiscale framework in order to preserve the proper near-wall behaviour while benefiting from a more structured multiscale modelling strategy. This illustrates an important point: even when WALE is not used in its original form, its operator remains attractive because it embeds a physically desirable near-wall asymptotic response. The same general philosophy also appears in later local closures such as the σ-model, which retains the wall-adapting strengths of WALE while addressing some of its more specific weaknesses (Nicoud *et al.* 2011).

The discussion of dynamic WALE models leads to a similar conclusion. Hybrid approaches suggest that the most promising future direction is not to discard WALE, but to combine its operator with additional mechanisms that improve adaptability and stability where needed (Toda *et al.* 2010).

From this perspective, WALE remains useful in two distinct senses. First, it is still a practical local SGS model for LES of wall-bounded turbulence, especially when correct near-wall behaviour and simplicity are important (Nicoud & Ducros 1999). Second, and perhaps more importantly, it continues to provide a conceptual foundation for later models. The enduring value of WALE therefore lies less in its status as a final answer to SGS modelling and more in the modelling principle it established: that a useful local SGS closure should respect near-wall physics from the outset. For that reason, WALE remains an important model in its own right and a foundation for more robust SGS formulations.

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